OpenShift Single Node (SNO) Deployment

1. Overview

This document describes the end-to-end procedure used to deploy a Single Node OpenShift (SNO) cluster in a disconnected environment using custom DNS, HAProxy, and a local mirror registry.

---

2. Environment Details

- Base Domain: `jaya.co`
- Cluster Name: `ocp`
- API Server: `api.ocp.jaya.co`
- Ingress: `*.apps.ocp.jaya.co`
- Network Plugin: `OVNKubernetes`

---

3. Install Configuration (install-config.yaml)

```
apiVersion: v1
baseDomain: jaya.co
metadata:
  name: ocp

compute:
- hyperthreading: Enabled
  name: worker
  replicas: 0

controlPlane:
  hyperthreading: Enabled
  name: master
  replicas: 1

networking:
  networkType: OVNKubernetes
```

```
  clusterNetwork:
  - cidr: 10.20.0.0/14
    hostPrefix: 23
  serviceNetwork:
  - 172.30.0.0/16

platform:
  none: {}

fips: false
pullSecret: ''
sshKey: ''
```

4. Pre-Requisites Setup

4.1 DNS Configuration (BIND)

Installed and configured BIND to resolve OpenShift cluster records:

```
 /etc/named.conf
options {
        listen-on port 53 { 127.0.0.1; 192.168.96.119; };
        listen-on-v6 port 53 { ::1; };

        directory "/var/named";

        dump-file "/var/named/data/cache_dump.db";
        statistics-file "/var/named/data/named_stats.txt";
        memstatistics-file "/var/named/data/named_mem_stats.txt";
        secroots-file "/var/named/data/named.secroots";
        recursing-file "/var/named/data/named.recursing";

        allow-query { localhost; 192.168.0.0/16; any; };
};

zone "ocp.jaya.co" IN {
        type master;
        file "forward.zone";
        allow-update { none; };
};
```

```
zone "96.168.192.in-addr.arpa" IN {
        type master;
        file "reverse.zone";
        allow-update { none; };
};

/var/named/forward.zone
$TTL 1D
@ IN SOA bastion.ocp.jaya.co. root.ocp.jaya.co. (
        2026041201
        1D
        1H
        1W
        3H )

@ IN NS bastion.ocp.jaya.co.

bastion   IN A 10.10.0.10
api       IN A 10.10.0.10
api-int   IN A 10.10.0.10
*.apps    IN A 10.10.0.10
master1   IN A 10.10.0.11

/var/named/reverse.zone
$TTL 1D
@ IN SOA bastion.ocp.jaya.co. root.ocp.jaya.co. (
        2026041201
        1D
        1H
        1W
        3H )

@ IN NS bastion.ocp.jaya.co.

10 IN PTR bastion.ocp.jaya.co.
11 IN PTR master1.ocp.jaya.co.
```

Required records:

- `api.ocp.jaya.co`
- `api-int.ocp.jaya.co`
- `*.apps.ocp.jaya.co`

Ensure proper forward and reverse resolution.

4.2 Load Balancer Configuration (HAProxy)

Configured HAProxy to route API and Ingress traffic:

```
/etc/haproxy/haproxy.cfg
global
  log         127.0.0.1 local2
  pidfile     /var/run/haproxy.pid
  maxconn     4000
  daemon

defaults
  mode tcp
  log global
  option dontlognull
  retries 3
  timeout connect 10s
  timeout client 1m
  timeout server 1m

frontend stats
  bind *:1936
  mode http
  stats enable
  stats uri /stats
  stats auth admin:jayaraj

# API Server
listen api-server-6443
  bind *:6443
  mode tcp
  server ocp-node 192.168.X.X:6443 check

# Machine Config Server (bootstrap phase only)
listen machine-config-server-22623
```

```
  bind *:22623
  mode tcp
  server ocp-node 192.168.X.X:22623 check

# Ingress 443
listen ingress-443
  bind *:443
  mode tcp
  server ocp-node 192.168.X.X:443 check

# Ingress 80
listen ingress-80
  bind *:80
  mode tcp
  server ocp-node 192.168.X.X:80 check
```

- Port 6443 → Kubernetes API
- Port 22623 → Machine Config Server
- Port 80/443 → Ingress router

4.3 Mirror Registry Setup

```yaml
kind: ImageSetConfiguration
apiVersion: mirror.openshift.io/v1alpha2
archiveSize: 4
storageConfig:
  registry:
    imageURL: bastion.ocp.task.lab:8443/mirror/oc-mirror-metadata
    skipTLS: false
mirror:
  platform:
    channels:
    - name: stable-4.18
      type: ocp
      minVersion: 4.18.23
      maxVersion: 4.18.23
    graph: true
```

5. Installation Workflow

Step 1: Create Install Manifests

Generate manifests:

```
openshift-install create manifests
```

Step 2: Create Ignition Configs

```
openshift-install create ignition-configs
```

Generated files:

- `bootstrap.ign`
- `master.ign`

Step 3: Host Ignition File

Installed HTTPD server and placed ignition files:

```
cp bootstrap.ign /var/www/html/
```

Step 4: Boot Node (SNO Master)

- Booted the master node using RHCOS ISO

```
coreosinstaller install /dev/sda -ignition-
url=http://192.168.96.119:8080/master.ign -insecure-ignition -copy-network
-insecure
```

6. Post Installation Validation

Check cluster health:

```
oc get nodes
oc get co
Oc get clusterversion
```

```
root@bastion:~/mirror
[root@bastion mirror]# oc get no
NAME                  STATUS   ROLES                         AGE    VERSION
master1.ocp.task.co   Ready    control-plane,master,worker   134m   v1.31.11
[root@bastion mirror]# oc get co
NAME                                       VERSION   AVAILABLE   PROGRESSING   DEGRADED   SINCE   MESSAGE
authentication                             4.18.23   True        False         False      99m
baremetal                                  4.18.23   True        False         False      126m
cloud-controller-manager                   4.18.23   True        False         False      128m
cloud-credential                           4.18.23   True        False         False      147m
cluster-autoscaler                         4.18.23   True        False         False      126m
config-operator                            4.18.23   True        False         False      126m
console                                    4.18.23   True        False         False      98m
control-plane-machine-set                  4.18.23   True        False         False      126m
csi-snapshot-controller                    4.18.23   True        False         False      126m
dns                                        4.18.23   True        False         False      125m
etcd                                       4.18.23   True        False         False      123m
image-registry                             4.18.23   True        False         False      115m
ingress                                    4.18.23   True        False         False      126m
insights                                   4.18.23   True        False         False      103m
kube-apiserver                             4.18.23   True        False         False      121m
kube-controller-manager                    4.18.23   True        False         False      124m
kube-scheduler                             4.18.23   True        False         False      123m
kube-storage-version-migrator              4.18.23   True        False         False      126m
machine-api                                4.18.23   True        False         False      126m
machine-approver                           4.18.23   True        False         False      126m
machine-config                             4.18.23   True        False         False      124m
marketplace                                4.18.23   True        False         False      125m
monitoring                                 4.18.23   True        False         False      99m
network                                    4.18.23   True        False         False      126m
node-tuning                                4.18.23   True        False         False      126m
olm                                        4.18.23   True        False         False      113m
openshift-apiserver                        4.18.23   True        False         False      106m
openshift-controller-manager               4.18.23   True        False         False      125m
openshift-samples                          4.18.23   True        False         False      115m
operator-lifecycle-manager                 4.18.23   True        False         False      125m
operator-lifecycle-manager-catalog         4.18.23   True        False         False      125m
operator-lifecycle-manager-packageserver   4.18.23   True        False         False      121m
service-ca                                 4.18.23   True        False         False      126m
storage                                    4.18.23   True        False         False      126m
[root@bastion mirror]# oc get clusterversion
NAME      VERSION   AVAILABLE   PROGRESSING   SINCE   STATUS
version   4.18.23   True        False         98m     Cluster version is 4.18.23
[root@bastion mirror]#
```

Expected result:

- Single control-plane node in Ready state
- All Cluster Operators Available=True